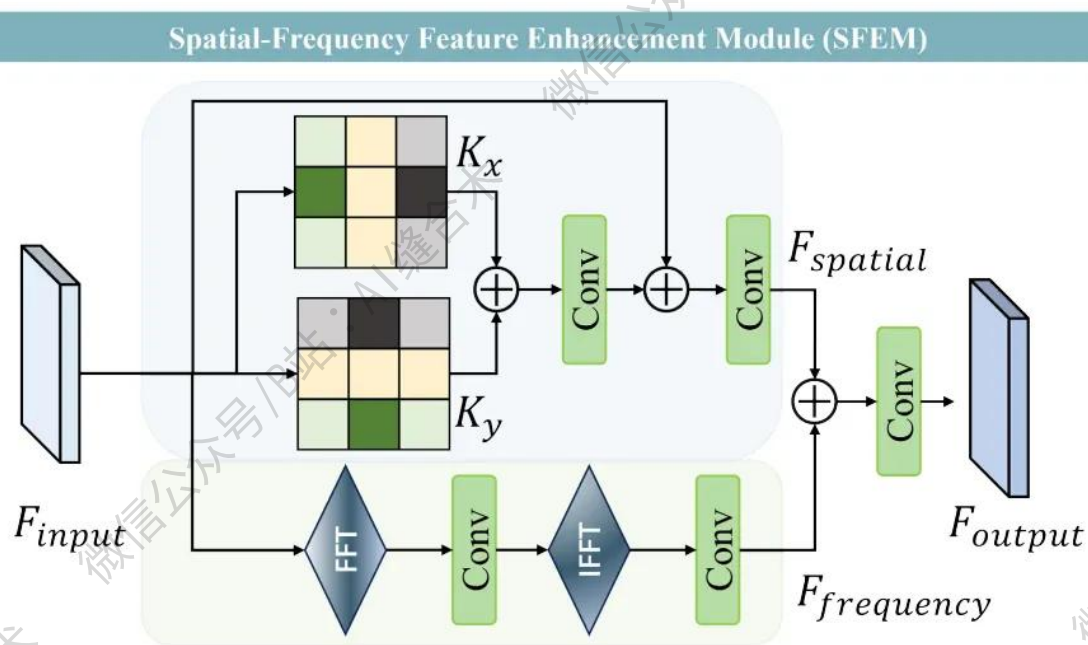


核心速览



论文题目: A Cross-Modality Feature Adaptive Interaction Approach for RGB-Infrared Object Detection in Aerial Imagery

中文题目: 一种用于航空影像中 RGB-红外目标检测的跨模态特征自适应交互方法

论文链接: <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11363305>

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核心速览:

本文提出了一种名为跨模态特征自适应检测 (CMFADet) 的新型框架, 用于解决航空影像中 RGB-红外 (IR) 目标检测面临的模态错位、特征融合退化和计算复杂度高等挑战, 通过**创新的空间-频率特征增强模块 (SFEM)**、

红外自适应特征聚合块（IR-AFAB）、通道交互融合（CIF）模块和自适应任务感知对齐头（ATAH），在多个数据集上实现了优于现有算法的性能。

<https://mp.weixin.qq.com/s/ZWI22oomVs-1hITSwkcAyQ>

```
SFEM(
  (sed): ScharrConv(
    (scharr_kernel_x_conv): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=64, bias=False)
    (scharr_kernel_y_conv): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), groups=64, bias=False)
  )
  (spatial_conv1): Conv(
    (conv): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (act): SiLU()
  )
  (spatial_conv2): Conv(
    (conv): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (act): SiLU()
  )
  (fft_conv): Conv(
    (conv): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
    (bn): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (act): SiLU()
  )
  (fft_conv2): Conv(
    (conv): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
    (bn): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (act): SiLU()
  )
  (final_conv): Conv(
    (conv): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
    (act): SiLU()
  )
)
input_tensor_shape : torch.Size([2, 64, 32, 32])
output_tensor_shape : torch.Size([2, 64, 32, 32])
```